Lab 8

(1) Recall the proof 3SAT is polynomial time mapping reducible to VERTEX-COVER ($3SAT \leq_P$ VERTEX-COVER). From $\phi = (x1 \vee x1 \vee x2) \wedge (\overline{x1} \vee \overline{x2} \vee \overline{x2}) \wedge (\overline{x1} \vee x2 \vee x2)$ construct the associated graph G. Explain how to obtain an 8-vertex-cover for G from a satisfying assignment for ϕ.

(2) Recall the proof 3SAT is polynomial time mapping reducible to CLIQUE ($3SAT \leq_P$ CLIQUE). From $\phi = (x1 \vee x1 \vee x2) \wedge (\overline{x1} \vee \overline{x2} \vee \overline{x2}) \wedge (\overline{x1} \vee x2 \vee x2)$ construct the associated graph G. Explain how to obtain a 3-clique for G from a satisfying assignment for ϕ.